

Cotton Field Log — the complete guide

Hi Emma. This replaces the spreadsheet, the R scripts and the macro button. There is nothing to install and nothing to code. You open a link, type numbers, and the Excel file is one click away whenever you or a mentor wants it.

Everything below is shown with the built-in sample trial. Load it yourself — **Setup → Load sample data** — and click around freely. Clearing it puts your real data back exactly as it was, so there is nothing you can break by exploring.

Cotton × *Microcystis* trial · Emma Doria

Live app: aidanjmeyers.github.io/ed-cotton-growth-dashboard

Generated August 25, 2026

The 60-second version

1. Open the link. Check **Logging as** (top right) says your name.
2. Type each plant's numbers on the **Log a day** tab. It saves itself.
3. On dosing days, press **Record a dose**.
4. When someone wants the data: **Data & export** → **Download Excel**.
5. When someone wants an update: **Status update** → **Copy to clipboard**.

Everything else in this guide is detail you can come back to.

1. Reading the top bar

Cotton × Microcystis TrialTAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25Sample dataLOGGING AS Emma

Log a dayPlants12WeatherData & exportWeekly updateFilesSetupHow to use

DATE

<08/25/2026

Today

Day 20 of the trial

TIME MEASURED

--:-- --

0/12
PLANTS LOGGED

Due today:HEIGHT DAILYLEAVES DAILYINTERNODE DAILYSPAD DAY 3 AFTER DOSELAI DAY 3 AFTER DOSE

2 days with nothing recorded:Tue, Aug 18Tue, Aug 11

HIGH	LOW	RAIN	HUMIDITY	CLOUD	MAX WIND	ET _a	DD60	DD60 SEASON
93.7 °F	78.2 °F	0 in	74 %	8 %	14 mph	0.242 in	26	556.7

Open-Meteo for 27.7151, -97.3275. DD60 = heat units that day, base 60°F; season total is cumulative from the trial start.

Last dose Sat, Aug 22 (3 days ago) · next fortnightly dose Sat, Sep 5

Record a dose

CONT-1CONTROL

Mon, Aug 24 (1 day ago): Height 25.4 · Leaves 7 · Internode 2.8

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

LOW-1LOW 2%

Mon, Aug 24 (1 day ago): Height 24 · Leaves 7 · Internode 2.5

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

HIGH-1HIGH 5%

Mon, Aug 24 (1 day ago): Height 22.2 · Leaves 6 · Internode 2.6

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

CONT-2CONTROL

Mon, Aug 24 (1 day ago): Height 25.7 · Leaves 7 · Internode 2.8

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

LOW-2LOW 2%

Mon, Aug 24 (1 day ago): Height 23.9 · Leaves 7 · Internode 2.5

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

HIGH-2HIGH 5%

Mon, Aug 24 (1 day ago): Height 21.3 · Leaves 6 · Internode 2.4

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

CONT-3CONTROL

Mon, Aug 24 (1 day ago): Height 26.3 · Leaves 7 · Internode 2.7

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

LOW-3LOW 2%

Mon, Aug 24 (1 day ago): Height 24 · Leaves 7 · Internode 2.8

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

HIGH-3HIGH 5%

Mon, Aug 24 (1 day ago): Height 21.3 · Leaves 7 · Internode 2.5

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

CONT-4CONTROL

Mon, Aug 24 (1 day ago): Height 25.7 · Leaves 7 · Internode 2.6

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

LOW-4LOW 2%

Mon, Aug 24 (1 day ago): Height 24.7 · Leaves 7 · Internode 2.5

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

HIGH-4HIGH 5%

Mon, Aug 24 (1 day ago): Height 22.2 · Leaves 6 · Internode 2.4

HEIGHT cm

LEAVES

INTERNODE cm

SPAD

LAI

note (optional)

historyclear

Day 20 · Tue, Aug 25 How many days into the trial today is

Saved & synced Everything you typed has reached the database. See [section 12](#) for the other things this can say

Logging as Your name goes onto every reading you record. Check it before you start

The tabs across the top are the eight places anything happens. They are covered in order below.

2. First time only — add your plants

Go to **Plants**.

Cotton × Microcystis TrialTAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25Sample dataLOGGING AS Emma

Log a dayPlants12WeatherData & exportWeekly updateFilesSetupHow to use

Add a plant

Everything here is set once, when the plant is started. The label is what you write on the pot.

LABEL

C-1

TREATMENT

Unassigned

REP

1

SOWN ON

08/25/2026

DEPTH (IN)

0.75

SEEDS

3

CONTAINER

5-gal

Add plant

Add a full set...

NOTES

anything about how this one was started

CONT-1CONTROL18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 25.4 · Leaves 7 · Internode 2.8

Open historyEditretire

LOW-1LOW 2%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 24 · Leaves 7 · Internode 2.5

Open historyEditretire

HIGH-1HIGH 5%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 22.2 · Leaves 6 · Internode 2.6

Open historyEditretire

CONT-2CONTROL18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 25.7 · Leaves 7 · Internode 2.8

Open historyEditretire

LOW-2LOW 2%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 23.9 · Leaves 7 · Internode 2.5

Open historyEditretire

HIGH-2HIGH 5%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 21.3 · Leaves 6 · Internode 2.4

Open historyEditretire

CONT-3CONTROL18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 26.3 · Leaves 7 · Internode 2.7

Open historyEditretire

LOW-3LOW 2%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 24 · Leaves 7 · Internode 2.8

Open historyEditretire

HIGH-3HIGH 5%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 21.3 · Leaves 7 · Internode 2.5

Open historyEditretire

CONT-4CONTROL18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 25.7 · Leaves 7 · Internode 2.6

Open historyEditretire

LOW-4LOW 2%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 24.7 · Leaves 7 · Internode 2.5

Open historyEditretire

HIGH-4HIGH 5%18 readings

Sown Wed, Aug 5 · day 20 · 5-gal · 0.75 in deep · 2 doses

Last reading Mon, Aug 24: Height 22.2 · Leaves 6 · Internode 2.4

Open historyEditretire

Add one plant with the label you wrote on the pot, its treatment group, when it was sown, how deep, and how many seeds.

Add a full set is faster when they all start together: fill in the shared details, type a prefix in **Label**, press **Add a full set...**, and say how many. It creates numbered plants split evenly across Control, Low 2% and High 5%.

Each plant tile then shows its age, container, sowing depth, how many doses it has had, its total readings, and its last reading.

- **Edit** — rename, change treatment, change notes
- **Retire** — for a plant that dies or leaves the trial. Its history is kept in full, it just stops appearing on the daily list. You can un-retire it

Adding a plant mid-trial is fine. So is retiring one.

3. The daily routine

Everything on the **Log a day** tab, top to bottom.

The date row

The date is already today. The arrows step one day at a time, **Today** jumps back. Backfilling an earlier day is completely normal — just pick the date.

Time measured is when you took the readings, not when you typed them.

12/12 plants logged tracks the day's progress.

Due today

The green band says what the schedule expects:

MEASUREMENT	WHEN
Height, leaves, internode	Every day
SPAD	Weekly, Fridays
Ceptometer (LAI)	Every other Friday
SPAD and LAI again	3 days after any dose

You never have to work the post-dose timing out yourself — the band says **DAY 3 AFTER DOSE** in orange when it applies. Boxes that are due are outlined in green.

Days with nothing recorded

Orange chips list days you missed. Tap one to jump straight there and fill it in.

The weather strip

Filled in for you from the plot's coordinates — you never fetch this.

DD60 is the heat units for that day; **DD60 SEASON** is the running total since the trial started. Cotton development tracks heat units better than calendar days, which is why it is there.

The plant cards

The grey line on each card is what that plant looked like last time, so you always know what you are comparing against.

Type the numbers. **There is no save button** — about a second after you stop typing the card turns green and says ✓ saved.

Under each box is a small arrow comparing what you just typed against that plant's last reading of the same thing: green up, red down. A red arrow is not an error. It is the plant telling you something.

Three time-savers:

- **Enter** jumps to the same box on the next plant, so you can go down the bench without touching the mouse
- **+ also record** reveals measurements that are not due today, if you took them anyway
- **history** > opens that plant's full record

Two rules that matter for the analysis

Blank is not zero. An empty box means you did not measure it. A zero means you measured and it was zero. Statistics treat those completely differently, so please leave things blank rather than typing 0.

Leaves is a total count — how many are on the plant right now, not how many are new. If it drops from 6 to 4 because two fell off, put 4. That fall is exactly the signal this trial is looking for.

The day note

At the bottom: one note for the whole day. Watering, dosing, weather you saw, anything odd. This is where "the AC in the greenhouse failed overnight" belongs.

4. Dosing days

Press **Record a dose** in the orange strip.

The screenshot shows the 'Cotton x Microcystis Trial' app interface. At the top, there's a header with the trial name, date (Day 20, Tue, Aug 25), and user (Emma). Below the header is a navigation bar with options like 'Log a day', 'Plants', 'Weather', etc. The main area displays various environmental data (HIGH, LOW, RAIN, HUMIDITY, CLOUD, MAX WIND, ET_a, DD60, DD60 SEASON) and a 'Record a dose' modal. The modal is titled 'Record a dose' and contains fields for DATE, VOLUME (ML), and STRENGTH %. It also includes a 'WORKS OUT AS' section with instructions on how to calculate the mixture (e.g., 2% = 2 mL stock + 98 mL water). Below this, there's a 'WHICH PLANTS' section with buttons for different plant groups (Control, Low 2%, High 5%, Everything, None) and specific plant IDs (CONT-1, LOW-1, HIGH-1, CONT-2, LOW-2, HIGH-2, CONT-3, LOW-3, HIGH-3, CONT-4, LOW-4, HIGH-4). A 'NOTE' field is also present. The modal has 'Cancel' and 'Record dose' buttons at the bottom.

Pick the date, then the plants — **clicking a group name selects that whole group**. Controls are left out by default.

Leave **Strength** blank and each plant is dosed at its own group's concentration. The panel spells out the actual mixture so you can make it up correctly:

2% = 2 mL stock + 98 mL water
5% = 5 mL stock + 95 mL water

Type a number in **Strength** only if you want to override every selected plant at once.

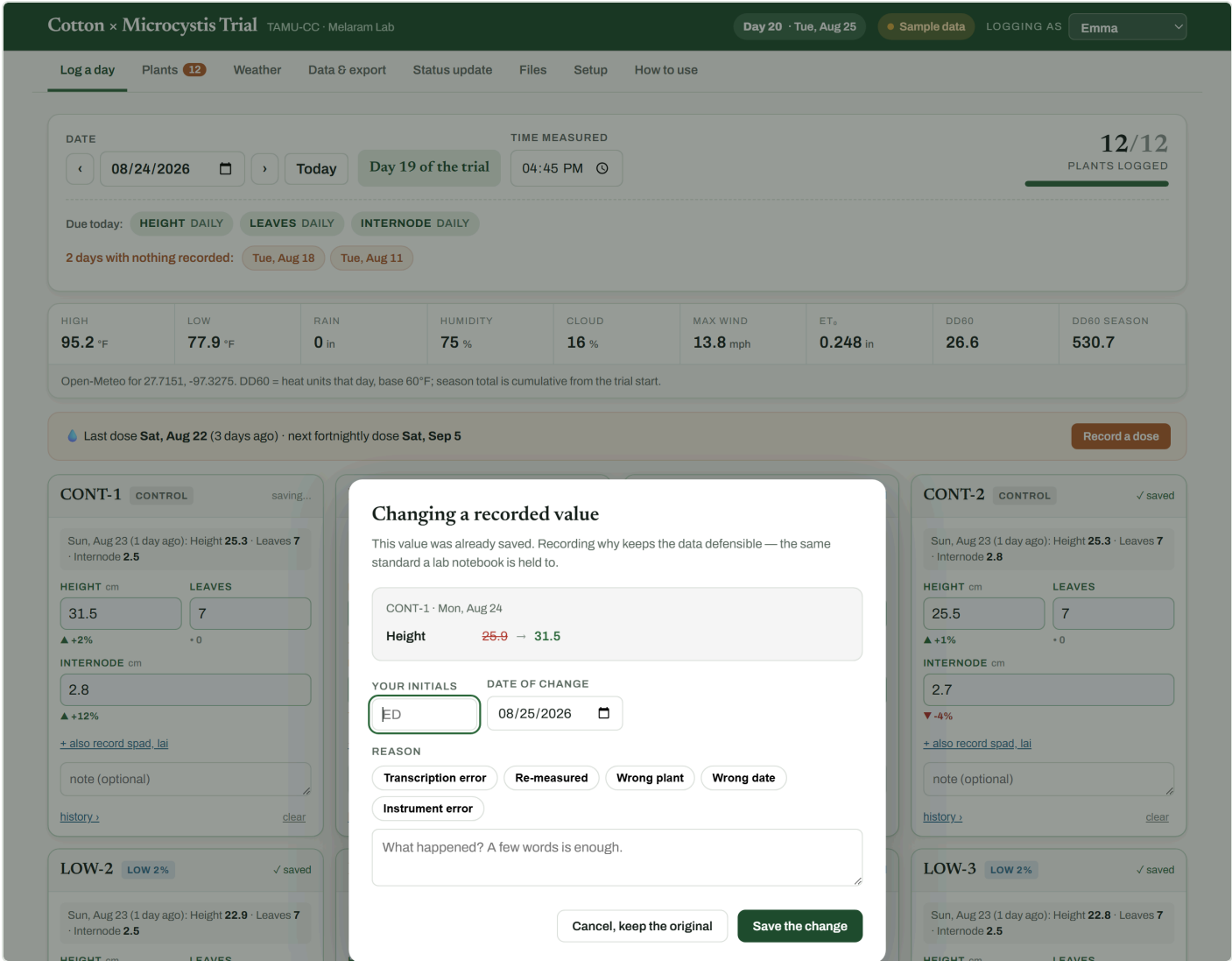
One record is written per plant, so each plant carries its own exposure history. From then on the app knows when the 3-day post-dose readings are due, every chart marks the dose day with a dashed line, and the strip tells you

when the next fortnightly dose falls.

5. Correcting something already saved

Type over a value, or press **clear** to remove a reading entirely.

If that value had already been committed — you saved it on an earlier visit, or someone else recorded it — this appears first:



Your initials	Remembered after the first time
Date of change	Defaults to today
Reason	Tap a common one, or type your own

It shows exactly what is about to change, and **nothing is saved until it is filled in**. **Cancel, keep the original** puts the value back.

Correcting something you typed a minute ago does *not* ask — that is just finishing your entry. It only asks once a reading has settled, and once a reading has been formally corrected it asks every time after that.

Why it matters: every number in your dataset either came straight off the instrument, or has your initials and a reason next to it explaining why it differs. If a committee asks "why is this point different from your raw sheet", the answer is in the file. The original value is never deleted.

Corrections appear on the plant's own page, on the **Corrections** sheet of the Excel export, and in the permanent record.

6. One plant at a time

Click any plant's name, anywhere.

← All plants

HIGH-1HIGH 5%

Sown Wed, Aug 5 · 20 days oldContainer 5-galDepth 0.75 inSeeds 3Rep 1Readings 18Doses 2

Edit plant

Log a reading today

HEIGHT
22.2 cm
▲ +4%
Aug 24

LEAVES
6
• 0
Aug 24

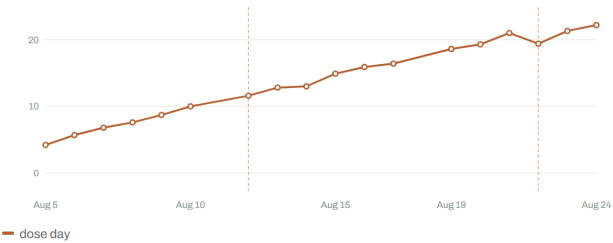
INTERNODE
2.6 cm
▲ +4%
Aug 24

SPAD
40.4
▲ +2%
Aug 21

LAI
1.1
▲ +39%
Aug 21

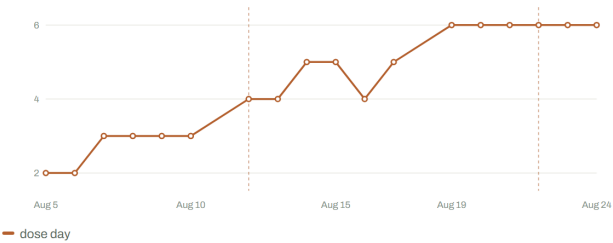
Height over time

cm · dashed lines are dose days



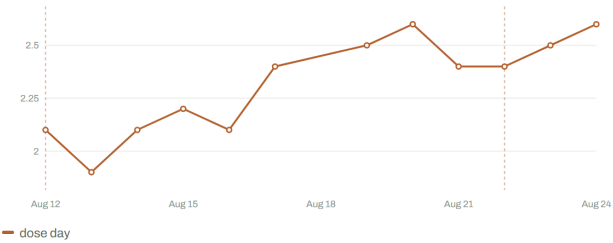
Leaves over time

count · dashed lines are dose days



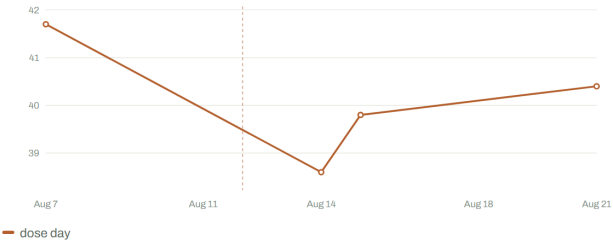
Internode over time

cm · dashed lines are dose days



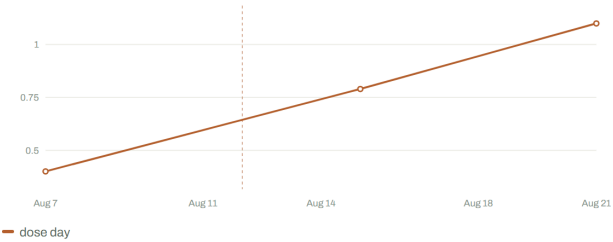
SPAD over time

SPAD value · dashed lines are dose days



LAI over time

leaf area index · dashed lines are dose days



Every reading

Arrows compare each value with this plant's previous reading of the same thing. Dose days are marked.

DATE	DAP	HEIGHT	LEAVES	INTERNODE	SPAD	LAI	DOSE	NOTE
2026-08-24	19	22.2 ▲+4%	6	2.6 ▲+4%	—	—	—	
2026-08-23	18	21.3 ▲+10%	6	2.5 ▲+4%	—	—	—	
2026-08-22	17	19.4 ▼-8%	6	2.4	—	—	5% · 100 ML	
2026-08-21	16	21 ▲+9%	6	2.4 ▼-8%	40.4 ▲+2%	1.1 ▲+39%	—	
2026-08-20	15	19.2 ▲+8%	6	2.5 ▲+8%	—	—	—	

ΣΥΝΟΛΟ ΣΥΝΟΛΟ	ΔΥ	ΔΥ ΔΥ ΔΥ ΔΥ	ΔΥ	ΔΥ ΔΥ ΔΥ ΔΥ	—	—	—
2026-08-19	14	18.6 ▲+13%	6 ▲+1	2.5 ▲+4%	—	—	—

You get its sowing details, its latest value for everything with the change since the reading before, a growth curve per measurement with dose days marked, every reading it has ever had with up/down arrows, and — if there are any — its corrections.

This is the screen to open when someone asks "how is the high-dose group doing", or when you want to see whether a plant turned the corner after an exposure.

7. Weather

Cotton × Microcystis TrialTAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25Sample dataLOGGING AS Emma

Log a dayPlants 12WeatherData & exportWeekly updateFilesSetupHow to use

DAYS OF WEATHER22

MEAN HIGH93.4 °F

MEAN LOW79.6 °F

HOTTEST95.9 °F

TOTAL RAIN0 in

HEAT UNITS DD60556.7

FROMmm/dd/yyyy

TOmm/dd/yyyy

Refresh weather

Weather to Excel

CSV

Temperature

daily high and low at the plot, °F

Aug 4Aug 9Aug 15Aug 20Aug 25

HighLowdose day

Rainfall

inches per day

No rain at the plot in this period.

Humidity and cloud

daily means, %

Aug 4Aug 9Aug 15Aug 20Aug 25

HumidityCloud

Cumulative heat units

DD60 since the trial start — the cotton growth clock

Aug 5Aug 10Aug 15Aug 20Aug 25

DATE	DAP	TEMP HIGH F	TEMP LOW F	TEMP MEAN F	RAIN IN	HUMIDITY %	CLOUD %	WIND MAX MPH	ET0 IN	SOLAR MJ/M2	DD60	DD60 CUMULATIVE	SOURCE
2026-08-25	20	93.7	78.2	85.4	0	74	8	14	0.242	26.1	26	556.7	open-meteo
2026-08-24	19	95.2	77.9	85.7	0	75	16	13.8	0.248	26.6	26.6	530.7	open-meteo
2026-08-23	18	94.9	76.4	85.1	0	76	8	11.1	0.241	26.4	25.7	504.1	open-meteo
2026-08-22	17	95.9	78.3	85.5	0	81	20	13.3	0.233	26.3	27.1	478.4	open-meteo
2026-08-21	16	93.7	80.6	85.7	0	82	37	15.5	0.226	26	27.2	451.3	open-meteo
2026-08-20	15	93.1	80	86.4	0	78	36	17.6	0.24	26.5	26.6	424.1	open-meteo
2026-08-19	14	92.4	79.8	85.4	0	79	47	15.4	0.235	26.4	26.1	397.5	open-meteo
2026-08-18	13	94.2	79.5	85.7	0	77	11	15.5	0.238	25.9	26.8	371.4	open-meteo
2026-08-17	12	92.2	78	85.3	0	81	17	13.5	0.223	25.8	25.1	344.6	open-meteo
2026-08-16	11	91.8	79.7	85.4	0	81	27	12.7	0.208	23.9	25.8	319.5	open-meteo
2026-08-15	10	94.4	81.1	86.3	0	79	28	15	0.223	24.2	27.8	293.7	open-meteo
2026-08-14	9	94	80.5	86	0	79	17	18.3	0.215	22.5	27.3	265.9	open-meteo
2026-08-13	8	93.8	80.8	86.6	0	77	22	19.1	0.252	26.7	27.3	238.6	open-meteo
2026-08-12	7	93.3	80.8	86.2	0	77	17	20.1	0.242	25.7	27.1	211.3	open-meteo
2026-08-11	6	93.3	80.6	86.8	0	78	26	16.6	0.234	25.2	26.9	184.2	open-meteo

Conditions at the plot for the whole trial: temperature, rainfall, humidity and cloud, and cumulative heat units. Filter by date, and export just the weather to Excel or CSV.

You never have to fetch this — it updates itself. **Refresh weather** only forces it. New days appear about a day late, which is normal for reanalysis data.

8. Data & export

Cotton × Microcystis TrialTAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25Sample dataLOGGING AS Emma

Log a dayPlants 12WeatherData & exportWeekly updateFilesSetupHow to use

PLANTS12

PLANT-DAYS RECORDED216

DAYS LOGGED18

DOSES LOGGED16

MEAN HEIGHT, LAST DAY23.9 cm

RAIN SINCE START0 in

HEAT UNITS DD60556.7

FROMmm/dd/yyyy

TOmm/dd/yyyy

PLANTAll plants

TREATMENTAll treatments

ROWSOne row per plant per day

Download Excel

CSV of this view

DATE	DAP	PLANT	TREATMENT	REP	PLANT AGE D	OBS TIME	HEIGHT (CM)	LEAVES	INTERNODE (CM)	SPAD	LAI	HEIGHT CHANGE	LEAVES CHANGE	INTERN
2026-08-05	0	CONT-1	Control	1	0	09:30	4.4	2	—	—	—	—	—	—
2026-08-05	0	CONT-2	Control	2	0	09:30	4.5	2	—	—	—	—	—	—
2026-08-05	0	CONT-3	Control	3	0	09:30	4.6	2	—	—	—	—	—	—
2026-08-05	0	CONT-4	Control	4	0	09:30	3.6	2	—	—	—	—	—	—
2026-08-05	0	HIGH-1	High 5%	1	0	09:30	4.2	2	—	—	—	—	—	—
2026-08-05	0	HIGH-2	High 5%	2	0	09:30	4.4	2	—	—	—	—	—	—
2026-08-05	0	HIGH-3	High 5%	3	0	09:30	3.7	2	—	—	—	—	—	—
2026-08-05	0	HIGH-4	High 5%	4	0	09:30	3.6	2	—	—	—	—	—	—
2026-08-05	0	LOW-1	Low 2%	1	0	09:30	4.4	2	—	—	—	—	—	—
2026-08-05	0	LOW-2	Low 2%	2	0	09:30	3.4	2	—	—	—	—	—	—
2026-08-05	0	LOW-3	Low 2%	3	0	09:30	3.8	2	—	—	—	—	—	—
2026-08-05	0	LOW-4	Low 2%	4	0	09:30	3.5	2	—	—	—	—	—	—
2026-08-06	1	CONT-1	Control	1	1	16:45	4.9	2	—	—	—	0.5	0	—
2026-08-06	1	CONT-2	Control	2	1	16:45	4.8	3	—	—	—	0.3	1	—
2026-08-06	1	CONT-3	Control	3	1	16:45	5.1	2	—	—	—	0.5	0	—
2026-08-06	1	CONT-4	Control	4	1	16:45	5.2	2	—	—	—	1.6	0	—
2026-08-06	1	HIGH-1	High 5%	1	1	16:45	5.7	2	—	—	—	1.5	0	—
2026-08-06	1	HIGH-2	High 5%	2	1	16:45	5.3	2	—	—	—	0.9	0	—
2026-08-06	1	HIGH-3	High 5%	3	1	16:45	4.9	2	—	—	—	1.2	0	—
2026-08-06	1	HIGH-4	High 5%	4	1	16:45	5.6	3	—	—	—	2	1	—
2026-08-06	1	LOW-1	Low 2%	1	1	16:45	4.6	2	—	—	—	0.2	0	—
2026-08-06	1	LOW-2	Low 2%	2	1	16:45	5.4	2	—	—	—	2	0	—

216 rows. Every growth row already carries that day's weather, so there is nothing to line up by hand.

The **Rows** dropdown switches what you are looking at:

VIEW	ONE ROW PER
Plant per day	Every individual reading — the raw data
Day	Whole-trial summary per day: means, standard deviations, counts
Treatment per day	Each group's mean and sd per day — the view to run stats from
Dosing log	Every exposure

Narrow it by date range, single plant, or treatment group. **CSV of this view** downloads exactly what is on screen.

Download Excel gives you one workbook with everything:

SHEET	WHAT'S ON IT
Growth	One row per plant per day, with change columns and that day's weather attached
Daily	Whole-trial summary per day
Treatment means	Per group per day — mean, sd, n
Doses	Every exposure: which plant, what strength, how much
Corrections	Every value changed after it was recorded, with initials and reason
Weather	Daily conditions and heat units
Plants	Your plant list with treatments and sowing details
Data dictionary	What every column means, in plain English

Because the weather is already on every growth row, there is no lining-up by hand and no VLOOKUP to write.

9. Status update

Cotton × Microcystis TrialTAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25Sample dataLOGGING AS Emma

Log a dayPlants12WeatherData & exportStatus updateFilesSetupHow to use

Status update

Where every plant stands right now, and how far it has moved. Pick a window and the whole summary rewrites itself — growth per group, what was dosed, what the weather did, and anything heading the wrong way. Copy it straight into an email.

WINDOW

REPORT BY

Last 7 daysLast 30 daysWhole trialCustom...Treatment group

Copy to clipboard.txt

Cotton × Microcystis Trial – status as of Tue, Aug 25

=====

Trial day 20 · 12 plants · TAMU-CC · Melaram Lab

Window: the last 7 days (Aug 19 – Aug 25)

WHERE THE PLANTS ARE NOW, BY TREATMENT GROUP

(current value, then the change across the last 7 days)

Control – 4 plants

Height:	25.7 cm	+5.6 (+28%)	last read Aug 24
Leaves:	7.3	+1.3	last read Aug 24
Internode:	2.8 cm	+0.3 (+12%)	last read Aug 24
SPAD:	43.3	-2.6 (-6%)	last read Aug 21
LAI:	1.2	+0.3 (+38%)	last read Aug 21

Low 2% – 4 plants

Height:	24 cm	+4.5 (+23%)	last read Aug 24
Leaves:	7	+1	last read Aug 24
Internode:	2.6 cm	+0.2 (+7%)	last read Aug 24
SPAD:	43.5	+0.8 (+2%)	last read Aug 21
LAI:	1.2	+0.4 (+41%)	last read Aug 21

High 5% – 4 plants

Height:	21.8 cm	+3.6 (+20%)	last read Aug 24
Leaves:	6.5	+0.8	last read Aug 24
Internode:	2.5 cm	+0.2 (+6%)	last read Aug 24
SPAD:	39.7	+1.3 (+3%)	last read Aug 21
LAI:	1.2	+0.3 (+39%)	last read Aug 21

COVERAGE

Readings on 6 of 7 days (Aug 19 – Aug 24)

72 plant-days recorded in this window

No readings: Aug 25

DOSING

Aug 22: 100 mL of 2% Microcystis to 4 plants (LOW-1, LOW-2, LOW-3, LOW-4)

Aug 22: 100 mL of 5% Microcystis to 4 plants (HIGH-1, HIGH-2, HIGH-3, HIGH-4)

CONDITIONS

Highs averaged 94.1°F (peak 95.9°F), lows 78.7°F

Rain 0 in over 0 wet day(s)

Heat units 185 DD60 this period; 556.7 DD60 since the trial started

Pick a window — **last 7 days**, **last 30 days**, **whole trial**, or custom — and the summary rewrites instantly:

High 5% – 4 plants

Height:	21.8 cm	+3.6 (+20%)	
Leaves:	6.5	+0.8	
SPAD:	39.7	+1.3 (+3%)	last read Aug 21

Then coverage, dosing, the weather, anything heading the wrong way, and your field notes. **Copy to clipboard** and paste it into an email.

A measurement that is not taken daily still shows its current value with the date it was last read, so a 7-day report is not full of blanks. And a 30-day window on a 20-day trial says so rather than pretending.

Report by switches between treatment groups and every plant individually.

10. Files — your working analysis

Drop your analysis workbook here whenever you update it. Anyone with the link can then see the live data *and* download your latest analysis, without you emailing anything. Protocols and photos are welcome too.

Label each upload (working file / protocol / photo / other) and add a note like "week 3, updated ANOVA".

Nothing is ever overwritten. Every upload is kept and tagged with your name and date, and the newest working file is marked latest so nobody grabs a stale one. All of them are copied into the permanent record twice a day.

11. Setup

Cotton × Microcystis TrialTAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25Sample dataLOGGING AS Emma

Log a dayPlants 12WeatherData & exportStatus updateFilesSetupHow to use

Study

Day 0 for the trial-wide DAP counter. Each plant also keeps its own sowing date.

TRIAL START DATE

08/05/2026

Save

Plot coordinates27.71514, -97.3275

Weather sourceOpen-Meteo (no key needed)

Weather days stored22

Refresh weather nowImport OpenWeather CSV

Measurement schedule

What the app expects on any given day. Change these in `config.js`.

Height (cm)	every day
Leaves (count)	every day
Internode (cm)	every day
SPAD (SPAD value)	weekly on Fri, plus 3 days after each dose
LAI (leaf area index)	every other Fri, plus 3 days after each dose

Where the data lives

On this device only. Data is saved in this browser — it survives closing the tab, but it does not follow you to another phone or computer, and file uploads are off. Ask Aidan to add the two Supabase values to `config.js` to switch that on.

Plants	12
Plant-days recorded	216
Doses logged	16
Waiting to sync	0

Sync nowDownload backupRestore backup

Permanent record

Twice a day everything is copied out of the database into files in the GitHub repo — CSVs, a full snapshot, dated backups, your uploaded analysis files, and the log of who changed what. Nothing here needs doing by hand.

Last snapshotnot available here

This panel reads a file that the published site has and a local copy does not. On the real link it shows when the data was last written to GitHub.

Check again

Practice mode

Loads a sample trial — 12 plants, three weeks of readings, two dosing rounds — so you can click around and try everything without touching real data. Clearing it puts things back exactly as they were.

Load sample dataClear sample data

Four panels. You will rarely need any of them, but here is what they do.

Study — the trial start date, the plot coordinates, and buttons to refresh the weather or import a weather file from the old R pipeline.

Measurement schedule — what the app expects and when. Changing it is a config edit, so ask Aidan.

Where the data lives — how many plants, readings and doses exist, and how many changes are waiting to sync. Also:

- **Sync now** — push anything waiting, when you have signal

- **Download backup** — saves the entire season to one file you can email
- **Restore backup** — loads one back in

Permanent record — when your data was last written to permanent files, how many changes have been logged, how many backups are kept, and who has been active this week. If that line turns red, the automatic backup has stopped — tell Aidan. It does not mean anything is lost.

Practice mode — loads a full sample trial to click around in. Clearing it restores your real data untouched.

12. If something looks wrong

The **pill at the top right** tells you the state of your data:

IT SAYS	MEANING
Saved & synced	Everything has reached the database. Normal
Offline — saved here	No signal. Your typing is safe on this device and will sync automatically
N waiting	That many changes still to send. Open the app on wifi, or press Sync now
On this device	Not connected to the shared database at all — tell Aidan
Sample data	You are in practice mode. Nothing you type is real

It works without signal. Anything typed in the greenhouse is held on the phone and syncs the moment you have bars again. Just do not clear your browser data while something is waiting.

Nothing is deleted by accident. The only removal is the `clear` link on a single card, and it asks for a reason first.

If a number looks wrong, there is a record of what it was before and who changed it.

If you are stuck, press **Setup** → **Download backup**, email that file to Aidan, and say what looked wrong. You cannot break this by clicking around.

13. Where your data actually lives

You do not have to do anything for this, but it is worth knowing.

Everything you type goes into a shared database immediately, which is why it appears on every device. Twice a day, all of it is also written into permanent files on GitHub: spreadsheets of every reading, dated backups, your uploaded analysis files, and a log of every change with who made it and when.

Practically: you cannot lose this data by dropping your phone.

The one habit that matters

Log every day you visit, even if it is a short visit and even if half the boxes are blank. Consistency is what makes the trends real. The orange "nothing recorded" chips exist to make catching up easy, not to nag you.